P8 Universe & Space

Physics | Classes 7 & 8 Combined

Classes 6 • 7 • 8 | TET Paper II Science

A · ASTRONOMY & UNIVERSE — Definitions & Units

Astronomy	Science that deals with the study of stars, planets, their motions, positions and compositions
Universe	Totality of everything that exists — consists of galaxies, planets, stars, meteorites, satellites and all matter/energy
Celestial Objects	Stars, planets, Moon and other objects like asteroids and comets in the sky
Solar System	The Sun and all celestial bodies revolving around it
Galaxy	Large collection of stars or cluster of stars and celestial bodies held together by gravitational attraction
Milky Way	Galaxy in which our Solar System is located — barred spiral galaxy
Exoplanets	Planets orbiting around stars other than our Sun — proves planetary systems exist throughout universe

Distance Units Used in Astronomy

Astronomical Unit (AU)	Average distance between Earth and Sun = 1 AU = 1.496×10^{8} km
Light Year (ly)	Distance light travels in one year = 1 ly = 9.4607×10^{12} km
Parsec (pc)	1 pc = 3.2615 ly = 3.09×10^{13} km — used for large cosmic distances
Neptune distance from Sun	30 AU — thirty times farther than Earth from Sun
Nearest star (Proxima Centauri)	4.22 light years from Earth and our Solar System
Earth from galactic centre	~27,000 light years away from the center of Milky Way

B · GEOCENTRIC & HELIOCENTRIC THEORIES

Theory	Core Claim	Proposed By	Evidence / Notes
Geocentric Theory (Geocentrism)	Earth is at the CENTRE of the Universe; Sun, Moon, stars and planets all orbit the Earth	Plato & Aristotle (6th century BC, Greece); standardized by Ptolemy (2nd century AD)	Based on: Sun appears to revolve around Earth daily; Earth feels stationary
Heliocentric Theory (Heliocentrism)	SUN is at the CENTRE; all planets including Earth revolve around the Sun	Nicolaus Copernicus proposed; Galileo Galilei confirmed with telescope observations (1610–11)	Galileo's Venus phase observations proved Venus orbits the Sun, not Earth

- Retrograde motion: apparent reversal of direction of planets — could NOT be explained by simple geocentric model
- Epicycle model: modification of geocentric model to explain retrograde — used by Ptolemy, Aryabhata, Tycho Brahe, Neelakanta Somayaji
- Telescope invented by Hans Lippershey; Galileo used it first to study the sky
- Galileo discovered: mountains on Moon, sunspots on Sun, dim stars invisible to naked eye, Milky Way is thousands of stars
- Venus phases (crescent to gibbous) → could only happen if Venus orbits the SUN → proved heliocentric model
- Aryabhata (Indian astronomer): said Earth spins on its axis — explains apparent E→W daily motion of celestial objects

C · ORIGIN OF UNIVERSE — Big Bang Theory

Big Bang Theory	Prevailing model of universe evolution — space and time emerged together ~14 billion years ago
Starting point	All matter was inside a bubble thousands of times smaller than a pinhead — hotter and denser than anything imaginable
Expansion	Suddenly expanded — grew from smaller than atom to bigger than galaxy in fraction of a second — still expanding today
First 3 minutes	Temperature dropped below 1 billion degrees Celsius
After 300,000 yrs	Universe cooled to ~3000°C — atomic nuclei captured electrons to form atoms — clouds of hydrogen and helium
After 100 M years	Gas became hot and dense enough for first stars to form — 10x higher star formation rate than today

Galaxy formation	Small galaxies formed ~1 billion years after Big Bang — closer together then — collisions created larger galaxies
Evidence	All galaxies are moving AWAY from us (redshift) — farther they are, faster they move — implies single origin point
Cosmic Microwave Background (CMB)	Faint glow in space — only direct evidence of the Big Bang itself

■ Big Bang: Space, Time and Matter all began together ~14 billion years ago. Universe is still expanding today.

D · GALAXIES — Types & Milky Way

Galaxy	Large collection of stars/celestial bodies held by gravity — billions of galaxies in universe
Size range	Most galaxies: thousands to ten thousand parsecs in diameter

Four Types of Galaxies

Type	Shape & Structure	Key Features	Example
Spiral Galaxy	Flat rotating disk with stars, gas and dust; central bulge (concentration of stars)	Spiral arms are sites of ongoing star formation — brighter due to young, hot stars	Many known spiral galaxies
Elliptical Galaxy	Three-dimensional ellipsoidal shape; smooth image; no spiral structure	Stars in random orbits; stars much OLDER than in spiral galaxies; surrounded by globular clusters	Common in galaxy clusters
Irregular Galaxy	No distinct regular shape — chaotic, no nuclear bulge, no spiral arms	About ONE-FOURTH of all galaxies; may have been spiral/elliptical deformed by external gravity; abundant gas and dust	~1/4 of known galaxies
Barred Spiral Galaxy	Spiral galaxy with central BAR-SHAPED structure composed of stars at center	Bars found in approximately 1/3 to 2/3 of all spiral galaxies	MILKY WAY is a barred spiral galaxy

Milky Way — Our Galaxy

Type	BARRED SPIRAL galaxy — our Solar System is located within it
Diameter	Over 100,000 light years

Number of stars	At least 100 billion (10^{11}) stars
Distance to centre	Our Solar System is ~27,000 light years from the galactic centre
Rotation speed	Solar system travels at ~828,000 km/h around Milky Way centre
Galactic year	Solar system takes ~230 million years to complete one orbit around Milky Way
Black hole	Monstrous black hole at centre — billions of times more massive than Sun
Nearest galaxy	Andromeda Galaxy (M31) — closest major galaxy to Milky Way
Indian name	"Akasha Ganga" — band of light in night sky
Resolved by	Galileo Galilei first resolved the Milky Way band into individual stars using telescope — 1610
Galaxies count	Edwin Hubble showed Milky Way is just ONE of many galaxies — early 1920s

E · STARS — Definition, Types & Key Facts

Star	Luminous heavenly body that radiates energy produced by nuclear fusion in its core
Why stars twinkle	Atmospheric disturbances bend light on its way to us — causes twinkling (stars are stationary)
Why planets don't twinkle	Planets are closer, appear as small discs — light from them is more stable
Visible to naked eye	~3,000 stars in night sky visible without telescope; ~2,000 commonly cited
Sun	The NEAREST star to Earth — average distance from Earth = 1 AU = 1.496×10^8 km
Nearest star system	Alpha Centauri — nearest star system to our Solar System; Proxima Centauri is 4.22 ly away

Moon Phases — Key Facts

- New Moon: Moon between Earth and Sun → illuminated side faces away from Earth → dark side toward Earth → Moon invisible
- Full Moon: Earth between Sun and Moon → illuminated side fully faces Earth → Moon appears round
- First Quarter (waxing half): Sun-Earth-Moon at 90° → half illuminated half visible

- Third Quarter (waning half): Moon at 90° on other side → half illuminated half visible
- Crescent: phases where Moon is LESS than half illuminated | Gibbous: MORE than half illuminated
- Waxing = growing/expanding illumination | Waning = shrinking/decreasing illumination
- Sangam literature (Purananuru 65) by Kalathalaiyar describes full moon observation — evidence of ancient Indian astronomy

F · CONSTELLATIONS — Definition, Key Constellations & Names

Constellation	Recognizable PATTERN of stars in the night sky when viewed from Earth — optical appearance, NOT a real physical system
Total number	International Astronomical Union (IAU) has classified 88 constellations to cover the entire celestial sphere
Names origin	Many have Greek or Latin names — often named after mythological characters
vs Galaxy	In galaxy: stars bound by gravity; in constellation: stars may be very far apart but appear close due to same direction
Why visible at different times	Different constellations visible at different times of year — due to Earth's revolution around the Sun

Key Constellations

Constellation	Indian Name	Key Feature / Stars
Ursa Major	Saptha Rishi Mandalam ([redacted])	Large constellation covering large sky area; 7 bright stars = Big Dipper (Sapta Maharishi in Indian astronomy)
Ursa Minor	Dhruva Mandalam	"Little bear" in Latin; lies in northern sky; contains POLE STAR — Polaris (Dhruva); 7 stars = Little Dipper
Orion	Vyadha / Kalpurusha	Named after hunter in Greek mythology; ~81 stars, only ~10 visible to naked eye; contains Betelgeuse, Rigel

Indian & English Constellation Names (Zodiac Signs)

Indian Name	English Name	Indian Name	English Name	Indian Name	English Name
Mesham	Aries	Rishabham	Taurus	Midhunam	Gemini
Kadakam	Cancer	Simmam	Leo	Kanni	Virgo
Thulam	Libra	Vrischikam	Scorpio	Dhanusu	Sagittarius
Makaram	Capricorn	Kumbam	Aquarius	Meenam	Pisces

G · SATELLITES — Natural & Artificial

Satellite	Object that revolves around a planet in a stable, consistent orbit
Natural Satellite	Natural objects revolving around a planet — also called MOONS — mostly spherical
Artificial Satellite	Man-made objects placed in orbit to rotate around a planet — usually Earth

Natural Satellites — Key Facts

- All planets EXCEPT Mercury and Venus have natural moons
- Earth has only ONE moon (The Moon) — average distance from Earth = 3,84,400 km | Diameter = 3,474 km
- Moon has NO atmosphere — does not produce its own light — reflects sunlight
- Jupiter and Saturn have more than 60 moons each
- Moon's rotation period = Revolution period around Earth → we always see the SAME side of Moon from Earth
- Some non-spherical moons = asteroids/meteors captured by strong planetary gravity

Artificial Satellites — World & India Firsts

World's first satellite	Sputnik-1 — launched by RUSSIA (USSR) — first artificial satellite in space
India's first satellite	ARYABHATA — built by ISRO, launched by Soviet Union on 19 April 1975 — named after Indian astronomer Aryabhata
First Indian satellite in Indian orbit	Rohini — launched in 1980 by Indian SLV-3 rocket — first satellite placed in orbit by Indian-made launch vehicle
Satellite uses	Television/radio transmission, weather forecasting, agriculture yield study, mineral resource location, GPS/navigation, Earth mapping

H · ISRO — Indian Space Research Organisation

Full form	Indian Space Research Organisation
Headquarters	Bangalore (Bengaluru), Karnataka
Formed	1969 — superseded INCOSPAR (Indian National Committee for Space Research) which was established in 1962
Founded by	Scientist Vikram Sarabhai — father of Indian space programme

Vision	"Harness space technology for national development while pursuing space science research and planetary exploration"
Managed by	Department of Space — reports to the Prime Minister of India
Popular rockets	PSLV (Polar Satellite Launch Vehicle) and GSLV (Geosynchronous Satellite Launch Vehicle)

ISRO Key Milestones

Aryabhata (1975)	19 April 1975 — India's first satellite — launched by Soviet Union
Rohini / SLV-3 (1980)	First satellite placed in orbit by an INDIAN-MADE launch vehicle — SLV-3 rocket
PSLV	Polar Satellite Launch Vehicle — for polar orbits
GSLV	Geosynchronous Satellite Launch Vehicle — for geostationary orbits
Cryogenic engine (2014)	January 2014 — GSLV-D5 used indigenous cryogenic engine to launch GSAT-14
20 satellites (2016)	18 June 2016 — ISRO launched 20 satellites in a single payload — set record then
104 satellites (2017)	15 February 2017 — PSLV-C37 launched 104 satellites in a SINGLE rocket — world record
GSLV-Mk III (2017)	5 June 2017 — India's heaviest rocket; placed GSAT-19 in orbit; capable of launching 4-ton satellites
GAGAN	GPS Aided Geo Augmented Navigation — satellite navigation system
IRNSS	Indian Regional Navigation Satellite System — India's own navigation system
Rakesh Sharma (1984)	2 April 1984 — First Indian to enter space — in joint India-USSR space programme; called "Cosmonaut"

I · ROCKETS — Parts, Propellants & Launch Principle

Rocket	Space vehicle with very powerful engine — carries people or equipment beyond Earth into space
First rockets	China invented rockets 800+ years ago — cardboard tube packed with gunpowder called "fire arrows" — used in 1232 AD
Rocket principle	Newton's Third Law — gas released DOWNWARD → rocket moves UPWARD — equal and opposite reaction

Multistage rocket	Single rocket cannot develop high velocities needed → multistage rockets used; upper stages ignite after lower stages exhaust

Four Parts of a Rocket

System	What It Does	Materials / Details
Structural System (Frame)	Frame that covers the rocket — provides shape and strength	Made of lightweight, strong materials like titanium or aluminium; fins at bottom for stability during flight
Payload System	Carries the satellite or cargo into orbit	Depends on mission — satellites for communication, weather, spying, planetary exploration, or manned missions
Guidance System	Guides and navigates the rocket in its correct path	Sensors, on-board computers, radars, communication equipment; tilts rocket to correct trajectory
Propulsion System	Provides thrust to move rocket — takes up most space in rocket	Fuel tanks, pumps, combustion chamber; two types: liquid and solid propulsion

Three Types of Propellants

Type	Description	Fuel Examples	Oxidizer Examples
Liquid Propellants	Fuel and oxidiser combined in combustion chamber; burn and exit from base with great force	Liquid hydrogen, Hydrazine, Ethyl alcohol	Oxygen, Ozone, Hydrogen peroxide, Fuming nitric acid
Solid Propellants	Fuel and oxidiser already combined — when ignited burn and produce heat; CANNOT be stopped once ignited	Polyurethanes, Poly butadienes	Nitrate salts, Chlorate salts
Cryogenic Propellants	Fuel/oxidiser or both are LIQUEFIED GASES stored at very LOW temperature; react on mixing — start own flame (no ignition system needed)	Liquid hydrogen (stored at -253°C)	Liquid oxygen (stored at -183°C)

J · CHANDRAYAAN MISSIONS — 1, 2 & 3

Chandrayaan-1 — Moon Vehicle

Detail	Information
Launch date	22 October 2008

Detail	Information
Launch vehicle	PSLV (Polar Satellite Launch Vehicle)
Launch site	Satish Dhawan Space Centre, Sriharikota, Andhra Pradesh
Lunar orbit insertion	8 November 2008
Orbit height	100 km from lunar surface
Duration	Operated for 312 days; achieved 95% of objectives
Communication lost	28 August 2009
Objectives	Find water on Moon; find elements of matter; search Helium-3; make 3D atlas of Moon; study solar system evolution
Key achievement	Discovered presence of WATER MOLECULES in lunar soil — major discovery
Other achievements	Confirmed Moon was completely molten once; 40,000+ images in 75 days; found caves on lunar surface; imaged Apollo-11 and Apollo-15 landing sites

Chandrayaan-2

- Launch date: 22 July 2019 by GSLV-Mk III rocket
- Consisted of: Orbiter + Lander (Vikram) + Rover (Pragyan — Sanskrit for "wisdom", 6-wheeled robotic vehicle)
- Vikram lander named after Dr. Vikram A. Sarabhai — father of Indian space programme
- Lunar orbit insertion: 20 August 2019
- Final stage: 7 September 2019 — Lander Vikram lost communication with ground station just 2.1 km above lunar surface — crashed
- Orbiter continues working successfully — still operational
- Project Director: Vanitha Muthayya — first woman project director at ISRO; born 2 August 1964 in Chennai

Chandrayaan-3 — Follow-up to Chandrayaan-2

- Launch date: 14 July 2023 from Satish Dhawan Space Centre, Sriharikota
- Lunar orbit insertion: 5 August 2023
- SUCCESSFUL landing: 23 August 2023 on lunar SOUTH POLE region
- India became FOURTH country to land on Moon (after Russia, USA, China) AND FIRST to land on lunar South Pole
- Consists of: Lander (Vikram) + Rover (Pragyan)

- Achievements: measured lunar soil temperature at various depths; detected moonquakes; Rover found Aluminium, Iron, Calcium, Sulphur, Oxygen on lunar surface
- Project Director: Dr. P. Veeramuthuvel — born 22 October 1976 in Villupuram district; PhD from IIT Madras; joined ISRO in 2014

K · MANGALYAAN — Mars Orbiter Mission (MOM)

Detail	Information
Mission name	Mars Orbiter Mission (MOM) — Mangalyaan means "Mars vehicle"
Launch date	5 November 2013
Launch vehicle	PSLV rocket from Sriharikota, Andhra Pradesh
Mars orbit insertion	24 September 2014
Significance	India's FIRST interplanetary mission — ISRO became 4th space agency to reach Mars (after Soviet Space Program, NASA, European Space Agency)
Historic first	India was FIRST NATION to succeed on its FIRST ATTEMPT to Mars | FIRST ASIAN country to reach Mars
Mission duration	Successfully completed 3+ years in Martian orbit; continued working as expected
Objectives	Develop interplanetary mission technology; explore Mars surface; study Mars atmosphere; investigate past/future life possibility

■ Mars — Key Facts

■ Fourth planet from Sun; second smallest planet in solar system

■ Called "Red Planet" — iron oxide (rust) on its surface and dusty atmosphere gives reddish colour

■ Mars rotation: once in 24 hours 37 minutes (similar to Earth's 24 hours)

■ Mars revolution: once in 687 days around Sun

■ Project Director of Chandrayaan-1: Dr. Mylsamy Annadurai — born 2 July 1958 in Kodhavadi near Pollachi, Coimbatore

L · NASA & INDIANS IN NASA

NASA	National Aeronautics and Space Administration
Headquarters	Washington, USA
Established	1st October 1958

Field centers	10 field centers — support NASA's work globally
International Space Station	NASA supports ISS — international collaborative space research project
Achievements	Landed rovers on Mars; analysed Jupiter atmosphere; explored Saturn and Mercury; satellites for weather, environment
NASA-ISRO partnership	NISAR Satellite — NASA-ISRO Synthetic Aperture Radar — joint programme

Apollo Missions — NASA's Most Famous Programme

- Apollo programme: total 17 missions — made American astronauts land on Moon

Apollo-8	First MANNED mission to go to the Moon — orbited Moon and returned to Earth
Apollo-11	First "Man Landing Mission" — landed on Moon on 20 July 1969
Neil Armstrong	First man to WALK on the surface of the Moon — Apollo-11
Crew of Apollo-11	Neil Armstrong, Buzz Aldrin (both landed on Moon), Michael Collins (stayed in orbit)

Indians in NASA

Person	Birth / Background	Key Achievement
Kalpana Chawla	Born 17 March 1962, Karnal, Punjab; joined NASA in 1988	First Indian woman astronaut in space (1997 Columbia Shuttle Mission); died in second mission (Columbia Shuttle) on re-entry; 10.4 million miles in 252 orbits, 372+ hours in space
Sunitha Williams	Born 19 September 1965 in USA; astronaut career started August 1998	Two trips to International Space Station; SET WORLD RECORD for longest spacewalk time by female astronaut — 50 hours 40 minutes total (7 spacewalks) in 2012; crew of NASA's Manned Mars Mission

M · Key Facts — Rapid Revision (Names, Dates, Records)

Scientists, Missions & Key Dates

Person / Event	Key Fact / Contribution	Date / Year
Plato & Aristotle	Proposed Geocentric model — Earth at center	6th century BC
Ptolemy	Standardized geocentric model (epicycle)	2nd century AD

Person / Event	Key Fact / Contribution	Date / Year
Aryabhata	Said Earth spins on its axis — Indian astronomer	5th–6th century AD
Nicolas Copernicus	Proposed Heliocentric model — Sun at center	16th century
Hans Lippershey	Invented the telescope	1608
Galileo Galilei	First used telescope to study sky; confirmed heliocentric using Venus phases	1610–11
Edwin Hubble	Showed Milky Way is just one of many galaxies	Early 1920s
Vikram Sarabhai	Founded INCOSPAR (1962); father of Indian space programme	1919–1971
ISRO founded	Indian Space Research Organisation — Bangalore	1969
Aryabhata satellite	India's first satellite — launched by Soviet Union	19 Apr 1975
Rohini / SLV-3	First Indian satellite by Indian launch vehicle	1980
Rakesh Sharma	First Indian in space — joint India-USSR programme	2 Apr 1984
Kalpana Chawla	First Indian woman in space (Columbia Shuttle)	1997
Chandrayaan-1 launch	PSLV from Sriharikota — to study Moon	22 Oct 2008
Chandrayaan-1 lunar orbit	Put into lunar orbit at 100 km height	8 Nov 2008
Chandrayaan-1 ended	Communication lost after 312 days; 95% objectives achieved	28 Aug 2009
Mangalyaan launch	PSLV from Sriharikota — India's first Mars mission	5 Nov 2013
Mangalyaan Mars orbit	India = 4th space agency, 1st Asian nation, 1st first-attempt success to Mars	24 Sep 2014
104 satellites record	PSLV-C37 — world record for satellites in single launch	15 Feb 2017
GSLV-Mk III first launch	India's heaviest rocket; placed GSAT-19; capable of 4-ton satellites	5 Jun 2017
Chandrayaan-2 launch	GSLV-Mk III; Orbiter+Vikram Lander+Pragyan Rover	22 Jul 2019
Chandrayaan-2 lunar orbit	Successfully inserted into lunar orbit	20 Aug 2019
Chandrayaan-2 crash	Vikram Lander lost contact 2.1 km above surface	7 Sep 2019

Person / Event	Key Fact / Contribution	Date / Year
Chandrayaan-3 launch	Follow-up mission to successfully land on Moon	14 Jul 2023
Chandrayaan-3 landing	Successfully landed on lunar SOUTH POLE — India 4th country to land on Moon	23 Aug 2023
NASA established	National Aeronautics and Space Administration — Washington, USA	1 Oct 1958
Apollo-8	First manned mission to go to Moon (orbit, no landing)	1968
Neil Armstrong	First human to walk on Moon — Apollo-11	20 Jul 1969
KalamSat	World's smallest satellite (64g) — built by high school students led by Rifath Sharook, 18, from Pallapatti, Karur district, TN	22 Jun 2017
Sunitha Williams	Longest spacewalk time by female astronaut — 50 hrs 40 min (7 walks)	2012
Dr. Annadurai (Chandrayaan-1)	Project Director — born in Kodhavadi near Pollachi, Coimbatore	1958–present
Vanitha Muthayya (CY-2)	Project Director Chandrayaan-2 — first woman PD at ISRO; born Chennai	1964–present
Dr. P. Veeramuthuvel (CY-3)	Project Director Chandrayaan-3 — born Villupuram; PhD from IIT Madras	1976–present
Dr. K.V. Sivan	ISRO Chairperson during Chandrayaan-2; born Sarakkalvilai, Kanyakumari; "Rocket Man"	in office 2018–2022
S. Chandrasekhar	Indian-American astrophysicist; Nobel Prize for Physics 1983; Chandrasekhar limit (white dwarf mass limit)	1910–1995

N · Glossary — Definitions

Astronomy — Science studying stars, planets, their motions, positions and compositions	**Universe** — Totality of everything that exists — all matter, energy, space and time
Celestial Object — Any natural object in space — stars, planets, Moon, asteroids, comets	**Solar System** — The Sun and all celestial bodies (planets, moons, asteroids) revolving around it
Galaxy — Large collection of stars/celestial bodies held together by gravitational attraction	**Milky Way** — Barred spiral galaxy containing our Solar System; 100+ billion stars; >100,000 ly diameter
Astronomy Unit (AU) — Average distance from Earth to Sun = 1.496×10^8 km	**Light Year (ly)** — Distance light travels in one year = 9.46×10^{12} km

Parsec (pc) — 1 pc = 3.2615 ly — unit for large cosmic distances	**Geocentric Theory** — Earth at center; all celestial bodies orbit Earth — Plato, Aristotle, Ptolemy
Heliocentric Theory — Sun at center; all planets including Earth orbit the Sun — Copernicus, confirmed by Galileo	**Retrograde Motion** — Apparent reversal of planet's direction of motion — key problem for geocentric model
Epicycle Model — Modified geocentric model to explain retrograde motion — used by Ptolemy and Aryabhata	**Big Bang Theory** — Prevailing model — universe began ~14 billion years ago from a single point; still expanding
Cosmic Microwave Background — Faint microwave glow — only direct evidence of the Big Bang	**Spiral Galaxy** — Flat rotating disk with spiral arms — sites of ongoing star formation
Elliptical Galaxy — 3D ellipsoidal shape; no structure; older stars; surrounded by globular clusters	**Irregular Galaxy** — No distinct shape; ~1/4 of all galaxies; may be deformed former spiral/elliptical
Barred Spiral Galaxy — Spiral with central bar-shaped structure — Milky Way is this type	**Star** — Luminous heavenly body that radiates energy from nuclear fusion in its core
Constellation — Recognizable pattern of stars in night sky; 88 total (IAU); optical appearance, not physical system	**Satellite** — Object revolving around a planet in stable orbit
Natural Satellite — Natural moon revolving around a planet — e.g., Earth's Moon	**Artificial Satellite** — Man-made object placed in orbit — e.g., Aryabhata, Chandrayaan
ISRO — Indian Space Research Organisation — Bangalore; formed 1969 — India's space agency	**Rocket** — Vehicle propelling itself by ejecting part of its mass — works on Newton's Third Law
Propellant — Chemical substance that undergoes combustion to produce thrust in rockets	**Liquid Propellant** — Fuel and oxidiser stored separately; combined in combustion chamber
Solid Propellant — Fuel and oxidiser pre-mixed; burns when ignited; cannot be stopped	**Cryogenic Propellant** — Liquefied gases stored at very low temperature; react on mixing; used in GSLV
PSLV — Polar Satellite Launch Vehicle — India's reliable rocket for polar orbits	**GSLV** — Geosynchronous Satellite Launch Vehicle — for geostationary orbits; uses cryogenic engine
Chandrayaan — Indian Moon missions — 1 (2008), 2 (2019), 3 (2023 — first landing on south pole)	**Mangalyaan** — Mars Orbiter Mission — India's first interplanetary mission; ISRO 4th space agency to reach Mars
NASA — National Aeronautics and Space Administration — USA's space agency; established 1 Oct 1958	**Apollo Mission** — NASA programme — 17 missions; Apollo-8 first to orbit Moon; Apollo-11 first Moon landing (1969)
Payload — Object/satellite carried into orbit by rocket; depends on mission	**Space Probe** — Unmanned vehicle sent into space to study planets — e.g., Mangalyaan
Exoplanet — Planet orbiting a star other than our Sun — proves planetary systems exist throughout universe	

■ Did You Know?

◆ KalamSat — world's smallest satellite weighing just 64 grams — built by high school students led by Rifath Sharook (18 years old) from Pallapatti near Karur, Tamil Nadu — launched by NASA on 22 June 2017.

◆ When Milky Way was last in the same position as today (~230 million years ago), there were no humans — not even Himalayan mountains — dinosaurs were roaming Earth.

◆ Subrahmanyan Chandrasekhar — Indian-American astrophysicist — won Nobel Prize for Physics in 1983 for his work on stellar evolution and black holes. Chandrasekhar limit: maximum mass (~1.4 solar masses) of a stable white dwarf star.

◆ India is the FIRST nation to successfully land on the lunar SOUTH POLE — with Chandrayaan-3 on 23 August 2023.

◆ Mars Orbiter Mission (Mangalyaan) cost only ■450 crore (~$74 million) — cheaper than making the Hollywood movie "Gravity" ($100 million) — making it the most cost-effective Mars mission in history.